airtel

SYNC

Stadium Surge

5G and 4G plan for a 1,00,000-fan final
₹3.52 Cr, sized from Airtel's site data

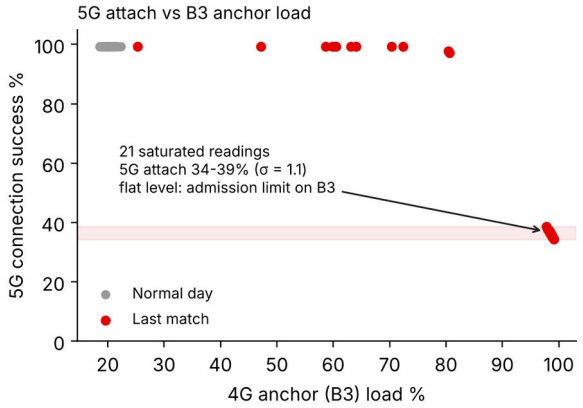
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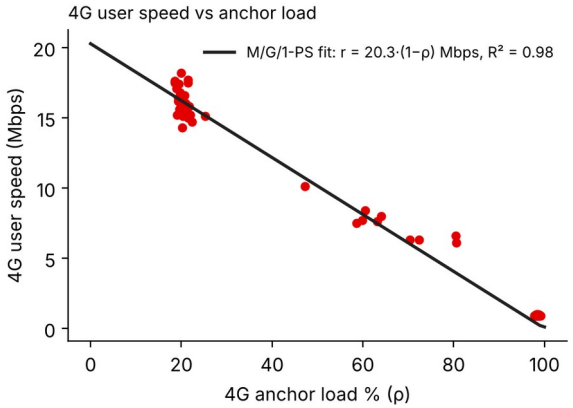


Platform Partner unstop

01 What broke at the last match

From Sync_V1.xlsx: 8 sites, 4 match moments, normal day vs the 65k-fan match





34-39% 5G attach success in all 21 saturated readings, spread of only 1.1

2.7x attach requests per available anchor slot at the innings break

R² 0.98 4G speed vs load fits r = 20.3(1-ρ) Mbps

24% expected 5G attach at the final if nothing changes

Reading the data

M6 stayed at 60% anchor load and kept 99.2% 5G attach. Every site above 97.8% dropped to roughly 36% and stayed there for three match phases. Plain congestion would scatter, so a level this flat points to an admission limit on B3 (RRC connected users).

Phones that could not add 5G stayed on LTE. That put about 2,100 extra users on a layer that was already full, which is why 4G fell to 1 Mbps.

02 Tasks 1 and 2: demand, cell count, budget

Cells sized at 80% load. 5G massive MIMO is loaded first since it gives the most Mbps per MHz

1,00,000	40,000	28k | 12k	5,040 | 2,160	25.2 | 6.5 Gbps
fans	Airtel (40%)	5G | 4G-only phones	online at peak (18%)	at 5 | 3 Mbps each

Cell type (DL)	Spectrum × SE	@80%	Users
5G macro 64T64R mMIMO	100 MHz × 74% × 16	947 Mbps	189 @5
5G small cell 4T4R	100 MHz × 74% × 7	414 Mbps	83 @5
4G macro sector	70 MHz × 2.5	140 Mbps	46 @3
4G small cell	40 MHz × 3.0	96 Mbps	32 @3

Requirement @80%	5G	4G
Peak demand	25.2 Gbps	6.5 Gbps
Cells needed (macro-eq.) / existing	27 / 12	47 / 20
Extra cells needed	**+15**	**+27**

Item	Qty	Cost	Adds
Under-seat small cells (4G + 5G)	24 × ₹8 L	₹1.92 Cr	9.9 Gbps 5G, 2.3 Gbps 4G
Cells on wheels (gates, M4/M6 side)	4 × ₹25 L	₹1.00 Cr	12 sectors each on 4G and 5G
Fibre ring, 2 × 10G	1.2 km	₹0.60 Cr	backhaul for new cells
Reserve: licences, RRC uplift, dry run	-	₹0.48 Cr	12% of budget
Total		**₹4.00 Cr**	

Capacity check @80% (Gbps)	5G	4G
Existing + 4 COWs + 24 small cells	32.7	6.8
Peak load at the final	**62%**	**76%**

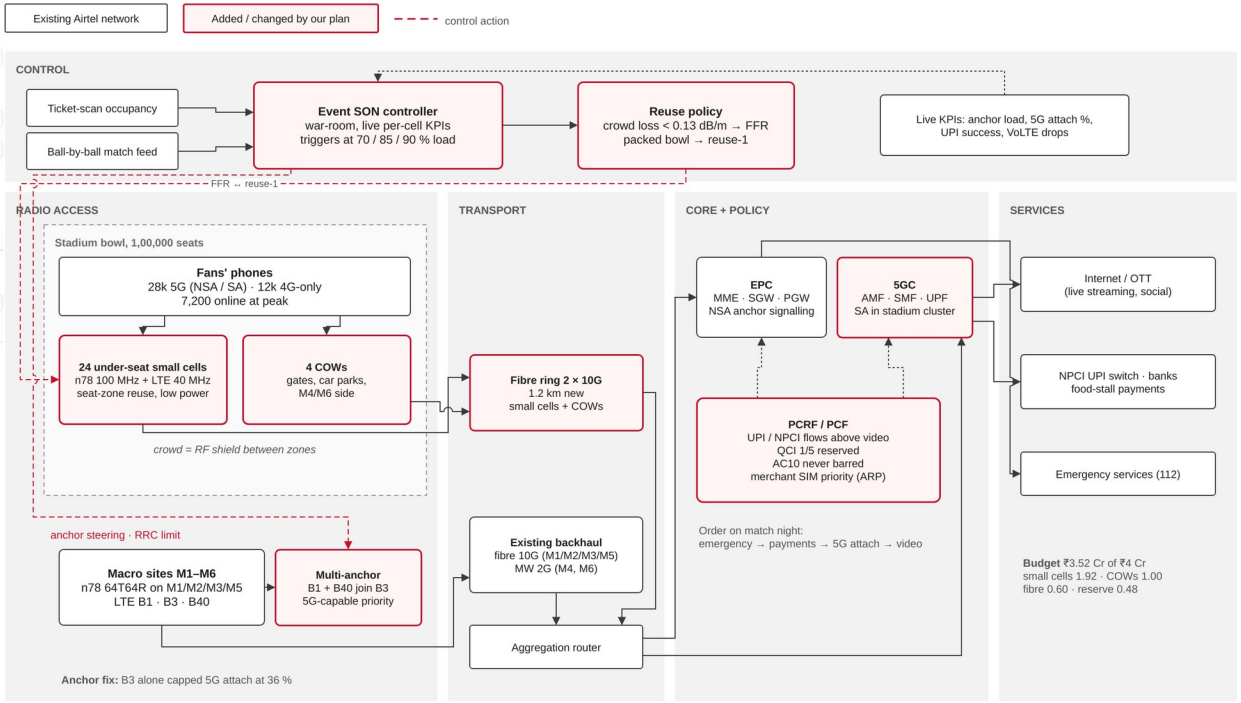
Budget note

Without the anchor changes, phones keep falling back at last match's rate and LTE demand reaches 16.2 Gbps. Covering that with COWs costs about ₹5.8 Cr. The small cells and fibre (₹2.52 Cr) stay for later matches. Only the COWs are a one-time cost.

03 Network changes for match night

Black outline: existing. Red outline: new or changed. Dashed red: control actions

Stadium Surge: network architecture for the final



How it fits together

- SON controller takes ticket-scan counts, the ball-by-ball feed and live cell KPIs
- Under-seat cells and COWs connect to a new 2 × 10G fibre ring
- B1 and B40 become NSA anchors next to B3
- SA core for the stadium cluster, PCF rules for UPI and 112

Code and demo

github.com/Amritha902/airtel-sync-stadium-surge

Demo video (65 s)

MATLAB, Simulink, Python, draw.io source

04 Task 3: 5G attach, payments, emergency calls

B3 ran out of RRC room, SgNB addition failed, and 5G phones stayed on LTE

Getting 5G phones onto 5G

- B1 and B40 as extra anchors
- Keep anchor capacity for EN-DC phones, move LTE-only phones to other bands (SPID/RFSP)
- Lower B1 threshold for SgNB add, no SgNB release on anchor load
- Higher RRC licence and a shorter inactivity timer on match day
- SA on n78 for phones that support it
- Target: 95%+ attach (was 36%)

UPI payments

- Small packets, but time critical. PCC rule for NPCI and bank endpoints, ahead of video
- Stall merchant SIMs on a higher ARP profile, small cells near the food courts
- Backup: UPI Lite, UPI 123PAY, stall Wi-Fi
- Target: under 1.5% failures (was 7.1%)

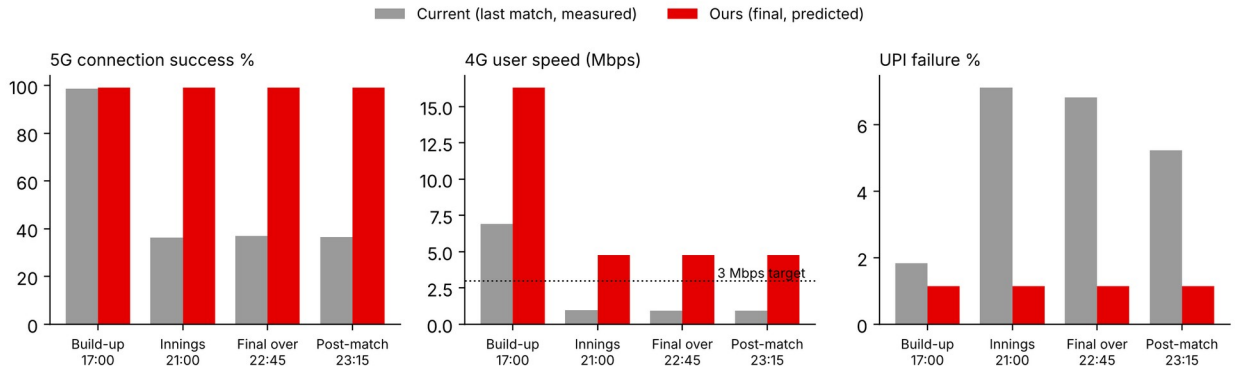
112 and VoLTE

- AC10 and QCI 1 never barred, pre-emption allowed
- About 5% PRB and RRC headroom kept for QCI 1/5
- AC 11-15 SIMs for police, medical and ground staff
- Barring for normal users only past 90% load
- Target: every emergency call sets up

Priority if a cell fills up: 112, then UPI, then 5G attach, then video.

05 Current vs proposed

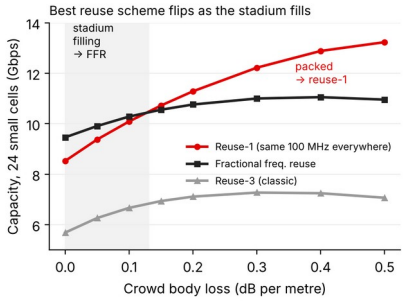
Grey: measured at the last match. Red: predicted for the final with the same fitted models. Below: our simulations

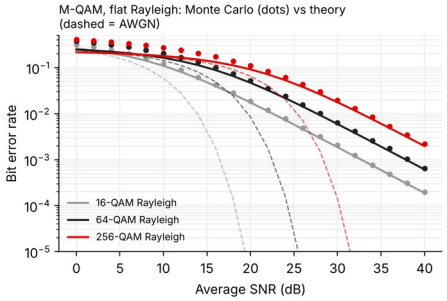


Seat-zone reuse

At 3.5 GHz a dense crowd adds body loss (TR 38.901 blockage model). Under-seat cells end up isolated from each other, so all 24 can run the full 100 MHz.

With the stands half empty FFR does better, so the controller changes scheme with occupancy.





Simulation results

- Full stadium, 24 cells: 12.2 Gbps with reuse-1, 11.0 with FFR, 7.3 with reuse-3
- Crossover near 0.13 dB/m crowd loss
- Rayleigh M-QAM Monte Carlo matches theory. 256-QAM needs about 25-30 dB
- 5 seeds, 5,040 users. Same code in MATLAB and Python

06 Task 4: user experience side effects

Mostly from adding 28 cells in and around one bowl

Possible impact	Cause	Mitigation
Interference, lower SINR	Many cells in a small area	Low-power under-seat antennas, downtilt, PCI/PRB planning for MU-MIMO
Handover ping-pong, dropped calls	Fans moving between macro, COW and small cell	Hysteresis and time-to-trigger tuning, MRO, checked at the dry run
Lower video quality for heavy users	Fair share and priority rules at peak	720p cap only when load crosses 85%
Short access delay	Barring for normal users	Only above 90% load, short barring timers
Coverage change for nearby homes	Macros re-tilted towards the stadium	Event tilt profile on 3 h before, reverted after
Post-match crush at gates and metro	Traffic moves from bowl to exits	2 COWs at the exits
Battery drain	NSA dual connectivity (4G + 5G)	SA on n78 drops the LTE leg, DRX tuning

Match-day monitoring

Per-cell dashboard (anchor load, 5G attach, UPI success, VoLTE drops) with actions at 70, 85 and 90% load. Dry run at a league match 30 days out, with rollback for every parameter. Extra capacity goes in 10 minutes before the innings break and the last over.

07 Competitive analysis and impact

Common ways operators handle big venues, judged against this case (60 days, ₹4 Cr)

	COW-heavy (common at Indian events)	Permanent DAS (large US venues)	5G SA only (Jio's network model)	Our plan
Fixes 5G attach failure	No	No	Yes, for SA phones	Yes: multi-anchor + SA
Capacity per seat	Medium, interference limited	High	Depends on radio layer	High: reuse-1 when packed
Ready in 60 days	Yes	No, months of civil work	Needs core changes	Yes
Cost profile	₹25 L every event	High capex	Core and device dependent	₹3.52 Cr, 73% reusable
Protects UPI and 112	Not addressed	Not addressed	Possible with slicing	PCC rules + barring policy
Sized from measured data	Rule of thumb	Venue survey	-	Fitted on Sync_V1 (R^2 0.98)

36% to 99%	1 to 4.8 Mbps	12 to 70+ Mbps	7.1% to 1.2%	all set up
5G attach	4G speed	5G speed	UPI failure	112 calls

Timeline: D-60 survey and orders | D-45 fibre and small cells | D-30 dry run at a league match | D-7 COWs placed | D+1 review

Within ₹4 Cr, about 40,000 Airtel fans get 5G that attaches, 4G above target, and payments that go through.

Code, MATLAB/Simulink and demo: github.com/Amritha902/airtel-sync-stadium-surge | demo video